

A Frozen Recoil–Branch Prediction for the Fine-Structure Constant

Pasquale Camelia
ORCID: 0009-0006-4779-4647

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Abstract

This note isolates, for preregistration, a single quantitative prediction of Quantum Gauge Theory (QGT): the values a photon-recoil determination of the fine-structure constant must return. The prediction is signed, dated ahead of the data that will test it, and falsifiable. It is extracted verbatim from the frozen section of the Second Edition (r26.9, `rd:s:frozen`) and from Register B of that volume’s Prediction Ledger (Appendix G); the underlying theory is not re-derived here, and none of it needs to hold for this note to be tested. What is claimed, what is admitted as input, what would confirm it, and what falls if it fails are all stated on this page. Its register, in the language of the volume, is a *conditional frozen forecast*: prospective and dated, but conditional on one discrete choice made *a posteriori* on existing data, and declared as such. It is not offered as an operatorially derived prediction, and it is not offered as proof of the theory.

1 The frozen values

Against the parameter-free fixed point $\alpha_{\text{fp}}^{-1} = 137.0359991703$, the frozen prediction for the two improved recoil determinations is

Determination	frozen prediction (4 July 2026)	current value
^{87}Rb recoil, improved	$\alpha^{-1} = 137.035999215$	137.035999206(11) [Morel 2020]
^{133}Cs recoil, improved	$\alpha^{-1} = 137.035999053$	137.035999046(27) [Parker 2018]
any new recoil species (Sr, Yb, . . .)	$w \in \{-0.7236, 0, +0.2764\}$	—

The recoil weight w_A of a species is defined by the apparatus contraction $\alpha_A^{-1} = \alpha_{\text{fp}}^{-1} + w_A f_1$, with the single-channel rate $f_1 = \varphi \cdot 10^{-7} = 1.618 \times 10^{-7}$. The blind content of the third row is its *discreteness*: a species not yet measured is forced onto three signed weights, not a continuum. No atomic datum enters the construction of the lattice itself.

2 Construction and admitted inputs

The values rest on three theorems of the volume meeting one hypothesis; the note states them only to expose the inputs, not to prove them.

- **Theorems (fixed structure, no atomic input).** The apparatus contraction $\alpha_A^{-1} = \alpha_{\text{fp}}^{-1} + w_A f_1$; the branch complementarity of the Schur return, giving the visible weight $Z_\varphi = \varphi/\sqrt{5} \approx 0.7236$ and the hidden weight $1 - Z_\varphi = 1/(\sqrt{5}\varphi) \approx 0.2764$; and the residue

reading of couplings. From these the branch displacements have exact closed forms in the golden ring,

$$\Delta_{\text{hidden}} = +\frac{10^{-7}}{\sqrt{5}}, \quad \Delta_{\text{visible}} = -\frac{\varphi^2 10^{-7}}{\sqrt{5}}, \quad \Delta_{\text{hidden}} - \Delta_{\text{visible}} = \varphi \cdot 10^{-7} = f_1.$$

- **Hypothesis.** A channel-weighted readout couples to *one* branch of the return, so admissible weights are the discrete lattice $w_A \in \{-Z_\varphi, 0, 1 - Z_\varphi\}$, with self-readouts at $w = 0$.
- **The one a-posteriori choice, declared.** Which measured channel reads which branch was assigned using the two existing recoil data: rubidium on the hidden branch, caesium on the visible branch with opposite orientation. Because that selection used the data, the 0.06σ agreement of the measured splitting with f_1 is part of the selection, *not* an independent check on past data. This is why the register is conditional, and it is the sharp line between this note and a derived prediction.

What is prospective, fixed with no further parameter once the branch pair is chosen: the splitting equals f_1 exactly (the branch weights differ by one), and the asymmetry is $|w_{\text{Cs}}|/|w_{\text{Rb}}| = Z_\varphi/(1 - Z_\varphi) = \varphi^2$. These stand as predictions on data yet to come.

3 Current status: survival, not confirmation

With the branch assignment above, both existing determinations fit: rubidium at 0.82σ on the hidden branch, caesium at 0.27σ on the visible branch. Among the golden assignments considered, each rival is dead beyond 2.3σ on at least one face — equipartition $\pm\frac{1}{2}$ at 4.1σ , the pair $(\varphi^{-2}, -\varphi^{-1})$ at 2.4σ , the pair $(0, -1)$ at 3.3σ , all on the rubidium face. Two disciplines are kept: the rival list is the list *considered*, not a predefined exhaustive space, so no global exclusion and no look-elsewhere correction is claimed; and, as above, the splitting agreement is inherited from the a-posteriori selection. The recoil pair therefore *survives*. Survival is not confirmation; the blind test is the next improved determination.

4 The decisive test, with thresholds

The prediction is signed, and the signs are its exposure. The true rubidium value is predicted to sit *above* the current central value, by $+9 \times 10^{-9}$ on α^{-1} ; the true caesium value likewise slightly above its current central value, by $+7 \times 10^{-9}$. The test becomes decisive at three standard deviations once a recoil determination reaches an uncertainty near 3×10^{-9} on α^{-1} — a relative precision of 2×10^{-11} , one further factor of four beyond the 2020 rubidium measurement, on the trajectory that programme is already on. No new apparatus class is required; the test is a precision improvement on an existing measurement.

5 Negative controls and the competing null

The competing null explanation is retained explicitly and is not treated as excluded: a conventional unrecognised systematic in one interferometer would reproduce a Rb–Cs offset without any of the present structure, and the metrological community holds exactly this possibility open. The reading here is *not* a spacetime variation of α : the invariant does not drift and only the channel contraction depends on the apparatus, so existing atomic-clock limits on a temporal drift of α are consistent with it and do not test it. The negative control for the branch structure is convergence: if improved determinations move together toward one value, the structure that this note predicts is absent. A ^{229}Th nuclear clock is deliberately *excluded* from the recoil

lattice: it reads a sensitivity $\delta\nu/\nu = K_\alpha \delta\alpha/\alpha + \dots$, not an absolute h/M recoil comparison, and placing it on the lattice would compare inhomogeneous observables. It enters elsewhere as a distinct, doubly conditional target (K_{Th} and w_{Th} to be derived first), named as a target and *not* predicted here.

6 Falsifiers, and what falls

The falsifiers are enumerated so that no retreat is available afterwards.

1. **Convergence.** If improved recoil determinations converge to a single common value well inside 10^{-7} , the channel-weighted class collapses and the tensorial reading of the coupling fails.
2. **Off-lattice weight.** If any new recoil weight lands off the lattice $\{-0.7236, 0, +0.2764\}$ at three standard deviations, the branch hypothesis fails even if the tensorial reading survives.
3. **Sign failure.** If the improved rubidium value drifts *away* from 137.035999215 at three standard deviations, the assignment fails in sign — the same death.

What falls if it fails is bounded and stated: a failure of (2) or (3) kills the branch assignment and this note, and leaves the rest of the volume standing; a failure of (1) is heavier — it removes the channel-weighted reading of α on which this note and the readout spectrum both rest. Neither outcome requires, or refutes, the remainder of QGT. That containment is the point of preregistering the recoil claim on its own.

Promotion path

One open step would turn this conditional forecast into a derived prediction: the channel-to-branch map from the measurement operator — a demonstration, from the contraction an h/M -against- R_∞ comparison actually performs, of *why* one recoil protocol reads the hidden branch and the other the visible branch with opposite orientation, sign included. That map is the single missing piece; the frozen numbers above do not wait for it.

Freeze and reproducibility

The values were constructed on 4 July 2026, ahead of any atomic-recoil or nuclear-clock determination of α published after that date, and are frozen. The machine-checkable record recomputes every frozen number from the volume’s own formulas (never hard-coded) and recomputes the current-data distances and rival exclusions from the primary 2018/2020 inputs.

Frozen record	<code>tools/qgt_branch_prediction_frozen.py</code>
Record SHA-256 (unchanged since r24.1)	<code>fb68dbbb869f5ade...56ce36c1</code>
r26.x source-tree SHA-256	<code>14a7a68be8aac900...b8220e08</code>

After deposit in an immutable archive with a DOI, the frozen register may change only to record a PASS, FAIL, or INCONCLUSIVE verdict of the data — never its numerical content.

Sources (QGT Second Edition r26.9): § “The frozen prediction: the branch lattice of the recoil weights” (`rd:s:frozen`, ch. 08); “The Prediction Ledger,” Register A and Register B (Appendix G); `FREEZE_MANIFEST.txt`.
Data: Morel et al., *Nature* **588**, 61 (2020) [Rb]; Parker et al., *Science* **360**, 191 (2018) [Cs].